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PALEOZOIC AND MESOZOIC PALEOMAGNETIC RESULTS FROM MOROCCO

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ABSTRACT

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A new, Lower Carboniferous geomagnetic pole at 4.8S,55.5E has been added for Morocco, and the Permian (31.6S,61.7E), Triassic (70.8S,67.4E), Jurassic (63.9S,39.5E), and Cretaceous (60.8S,61.9E) for Africa modified by the incorporation of new data. The Lower Carboniferous pole requires a re-interpretation of the Carboniferous—Permian path. The Mesozoic geomagnetic poles while not significantly different line up to suggest a small loop.

INTRODUCTION

Morocco, at the northwestern extremity of Africa, contains within its borders a considerable variety of rocks and geological structures. The relationship of the Hercynian belt to the West African (Saharan) craton may be determined, and this may then be compared to sequences known primarily from southern Africa. In addition, intercontinental comparisons throw light on the relation of Africa with America and the Mesozoic disruption of Gondwana. Finally, there is the prospect of augmenting the number of results from the Lower Paleozoic and Precambrian, the latter bearing on questions concerning earlier phases of drift (Hurley and Rand, 1969; Piper, 1973; Spall, 1972; Nairn, 1975; and Briden, 1967b).

The first published paleomagnetic results of Helsley (1965) were followed by Tarling and Sutton (1967), Hailwood (1972, 1974, 1975), Hailwood and Mitchell (1971), and Bardon et al. (1973a,b). The most extensive survey is that of Hailwood (1972); the others have in general sampled from a restrict-

ed suite of rocks, a very heavy emphasis having been laid upon Upper Triassic lavas and sediments, and Lower Jurassic igneous rocks from which reliable results were obtained. Work on younger Mesozoic and Cenozoic rocks by French and German groups is still in progress.

The study presents some new results from Africa, including Carboniferous and Permian pole positions which modify the African polar-wandering curve, and results from younger rocks confirming those already obtained, and summarizes the present status of paleomagnetic work on Paleozoic and Mesozoic rocks from Morocco (see also Martin, 1976).

TECHNIQUES AND SAMPLING

Oriented block samples and/or cores were collected at a number of sites in the High Atlas, Anti-Atlas, Moroccan Meseta, and the Sahara Platform (Fig. 1). The rocks ranged in age from Cambrian to Cretaceous. Most were

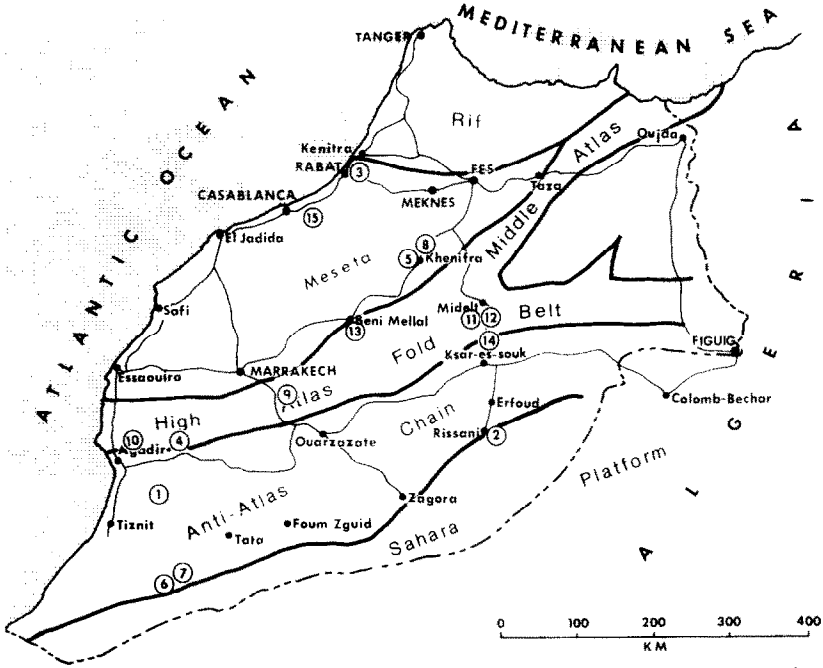


Fig. 1. The principal physiographic/tectonic units of Morocco. Main roads and towns are shown for orientation. The numbers refer to general sampling locations identified in the text. Legend: *1* = Cambrian near Souk Ait Baha el Arba, *2* = Devonian southeast of Rissani, *3* = Carboniferous flow of Meseta, *4* = Tizi-n-Tichka massif, *5* = Djebel Hadid redbeds, Khenifra, *6* = Visean limestones, Tuisgui, *7* = Aftez Visean Limestone, Oued Draa, *8* = Permian north of Khenifra, *9* = Lava near Telouet, *10* = Argana redbeds near Agadir, *11* = Imiter massif south of Midelt, *12* = Limestone turbidites south of Midelt, *13* = Liassic carbonates south of Beni Mellal, *14* = Infra-Cenomanian north of Ksar-es-Souk.

dated by normal stratigraphic and paleontological techniques with sampling concentrated in regions where re-mapping by members of the University of South Carolina geological survey project was in progress. Certain of the igneous rocks were dated radiometrically.

Remanence measurements on standard-sized cylinders were made either by using a Schonstedt SSM-1A or a Digico Complete Results magnetometer. Magnetic cleaning along three orthogonal axes made use of a Schonstedt GSD-1 alternating field demagnetizing unit, while thermal cleaning was carried out within a mumetal screened room in which the residual field was reduced to less than 15 gamma by ancillary coils. The technique of progressive demagnetization of pilot samples at relatively close intervals (usually 50 Oe or 50°C) was employed. Stable end-point curves (Zijderveld, 1967) provided a clear illustration of the direction of magnetization of the hard magnetic component (Fig. 2a—g). This component is usually regarded as primary, but there are reasons for considering that it may on occasion, either be secondary or the resultant of more than one stable component (see discussion in Tarling and Sutton, 1967; Larson and Walker, 1975; and Roy and Park, 1974). After determination of the optimum demagnetization stage, the remaining samples were usually demagnetized in three fields to include measurements on each side of the optimum. The values recorded in the tables were then the result at that stage at which the scatter and confidence ovals were minimal. In those rocks where no stable end point could be reached further work was abandoned.

Cambrian—Precambrian

Samples were collected from the Serie lie de vin on the northwest-facing scarp of the Anti-Atlas southwest of Souk Ait Baha el Arba (Fig. 1, *1*). Stratigraphically the Serie lie de vin represents the deposits formed during a regressive phase succeeding the limestones and dolomites of the Adoudounian. The Adoudounian considered by most as Infra-Cambrian, may be lowest Cambrian, thus, there is still some doubt as to the exact stratigraphical location of the Serie lie de vin.

Samples were thermally demagnetized and showed a minimum dispersion at 450°C (Fig. 3a). The mean direction and other parameters at this stage of treatment are recorded in Table I.

Devonian

Although different lithological types were sampled (a syenite in the Bou Regreg near Rabat, a granite near Tiflet, and a suite of reef limestone samples from southeast of Rissani (Fig. 1, 2), we were not able to obtain a stable end-point curve in which declination and inclination decreased towards the origin. The directions of magnetization after the cleaning attempts on the igneous rocks still remained grouped about the present field



Fig. 2. Stable end-point curves. Normalized intensity is measured radially from the origin. The figure shows declination in plan view and inclination in meridional section. Steps are shown in oersteds (a, c, e) or in °C (b, d, f). a. Thermal demagnetization of sample from the Serie lie de vin. b. Alternating field demagnetization of Carboniferous lava sample. c. Thermal demagnetization of Visean redbed sample. d. Alternating field demagnetization of Carboniferous limestone sample. e. Thermal demagnetization of Permian redbed sample. f. Alternating field demagnetization of Liassic limestone turbidites. g. Thermal demagnetization of Infra-Cenomanian sandstone sample.

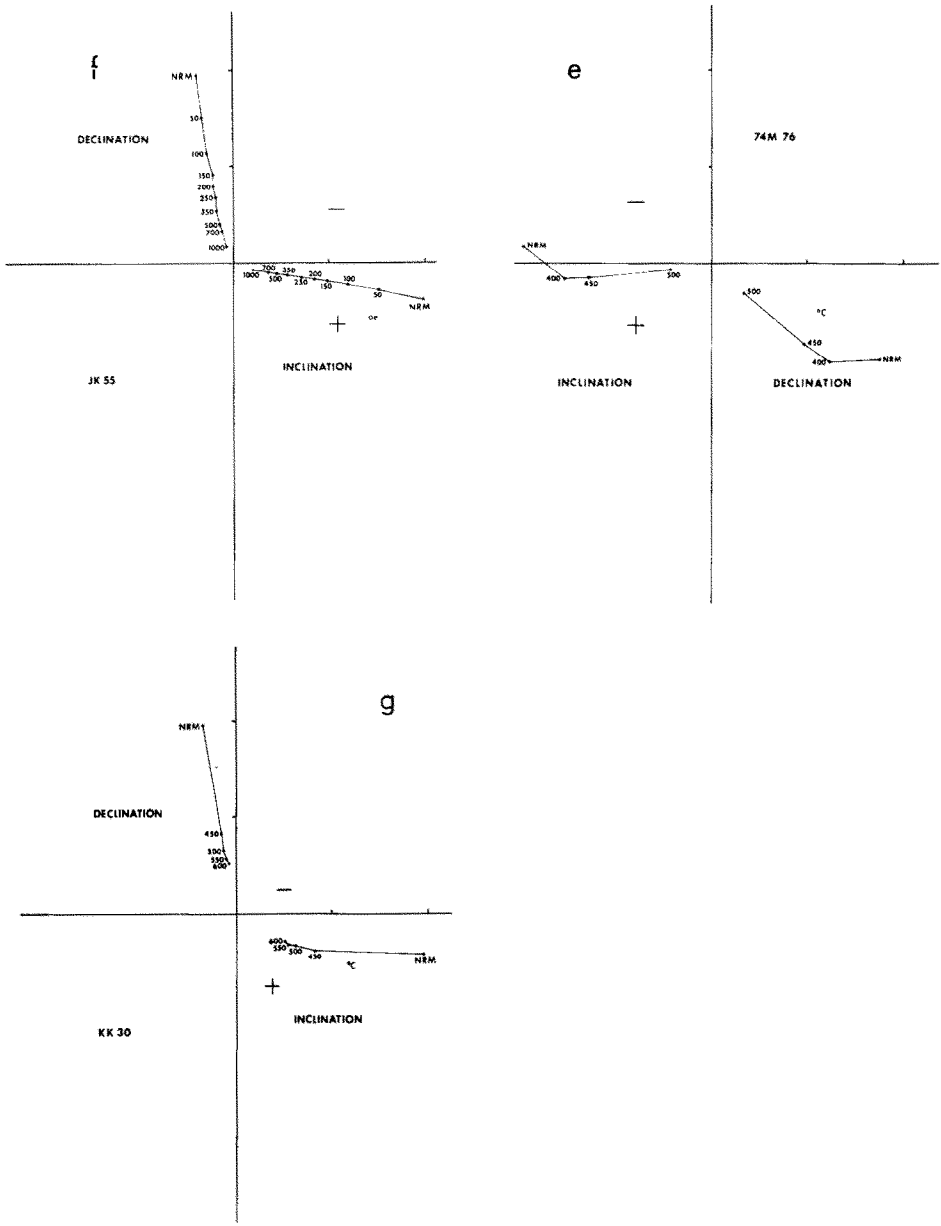


Fig. 2. f, e and g.

direction. A geomagnetic pole calculated for the single suite of limestones was essentially the same as that obtained for Permian rocks, suggesting that the limestones carry a stable magnetization impressed during Permian times. These data are therefore not considered further.

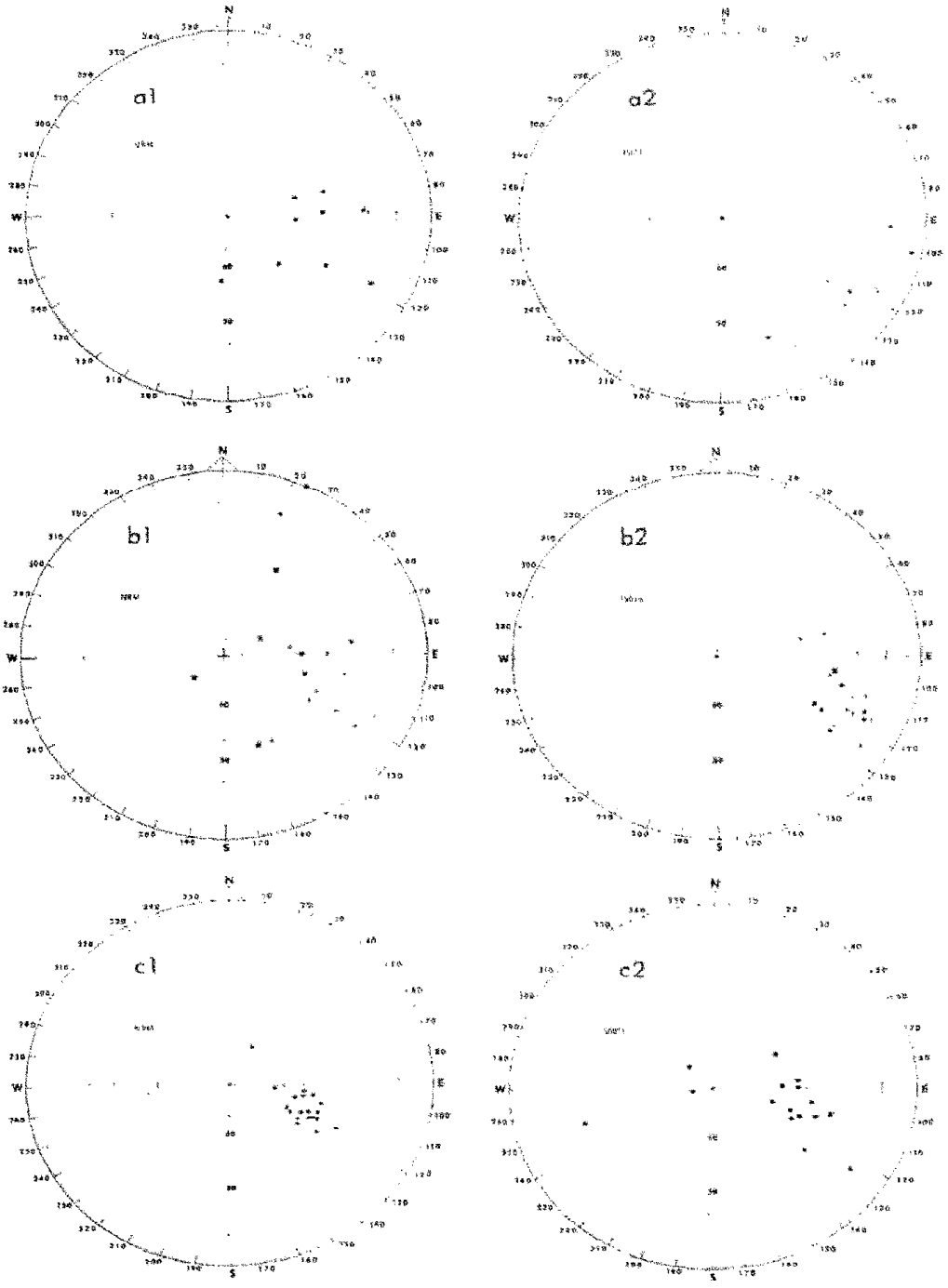


Fig. 3. a, b and c. For legend see p. 98.

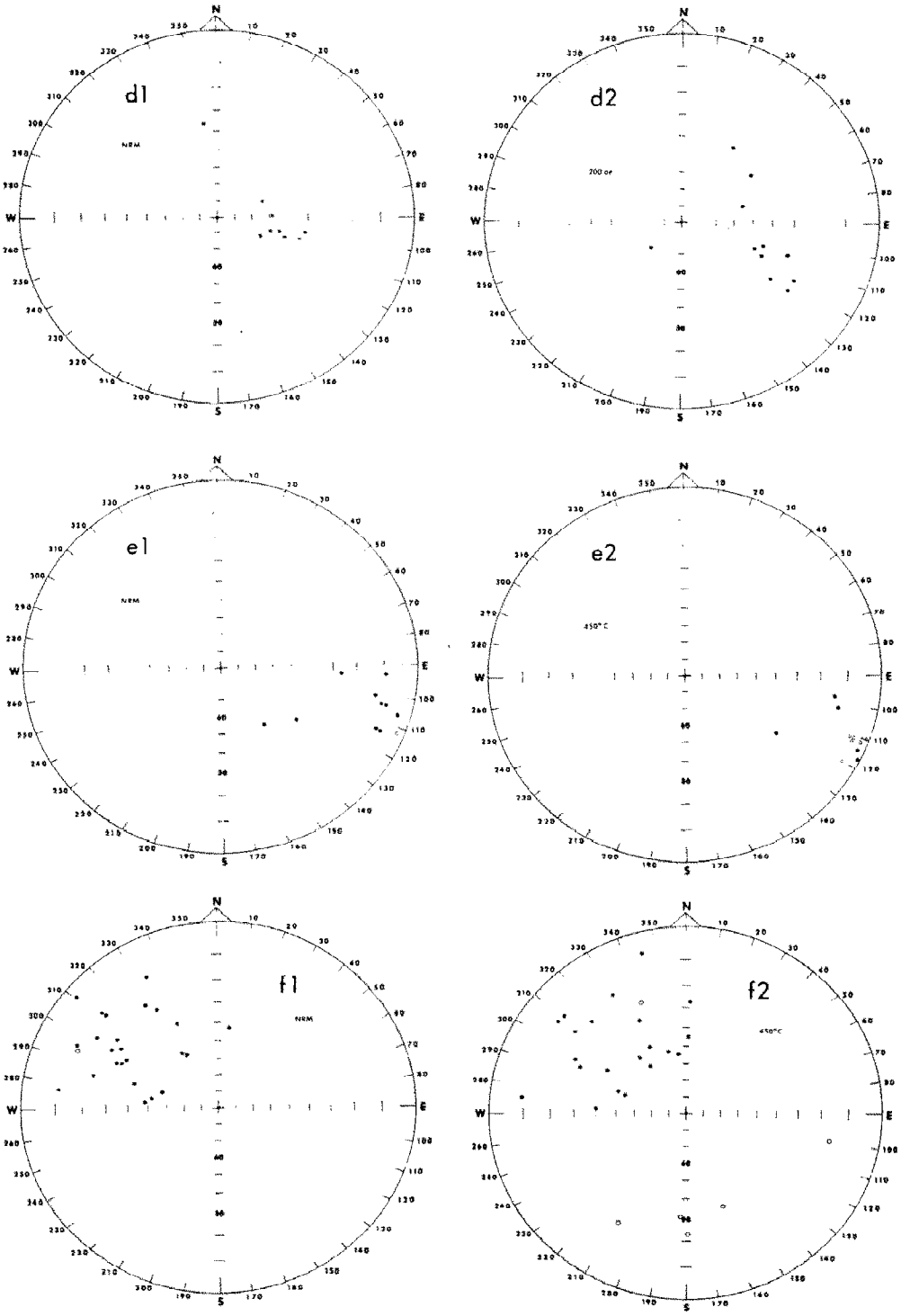


Fig. 3. d, e and f. For legend see p. 98.

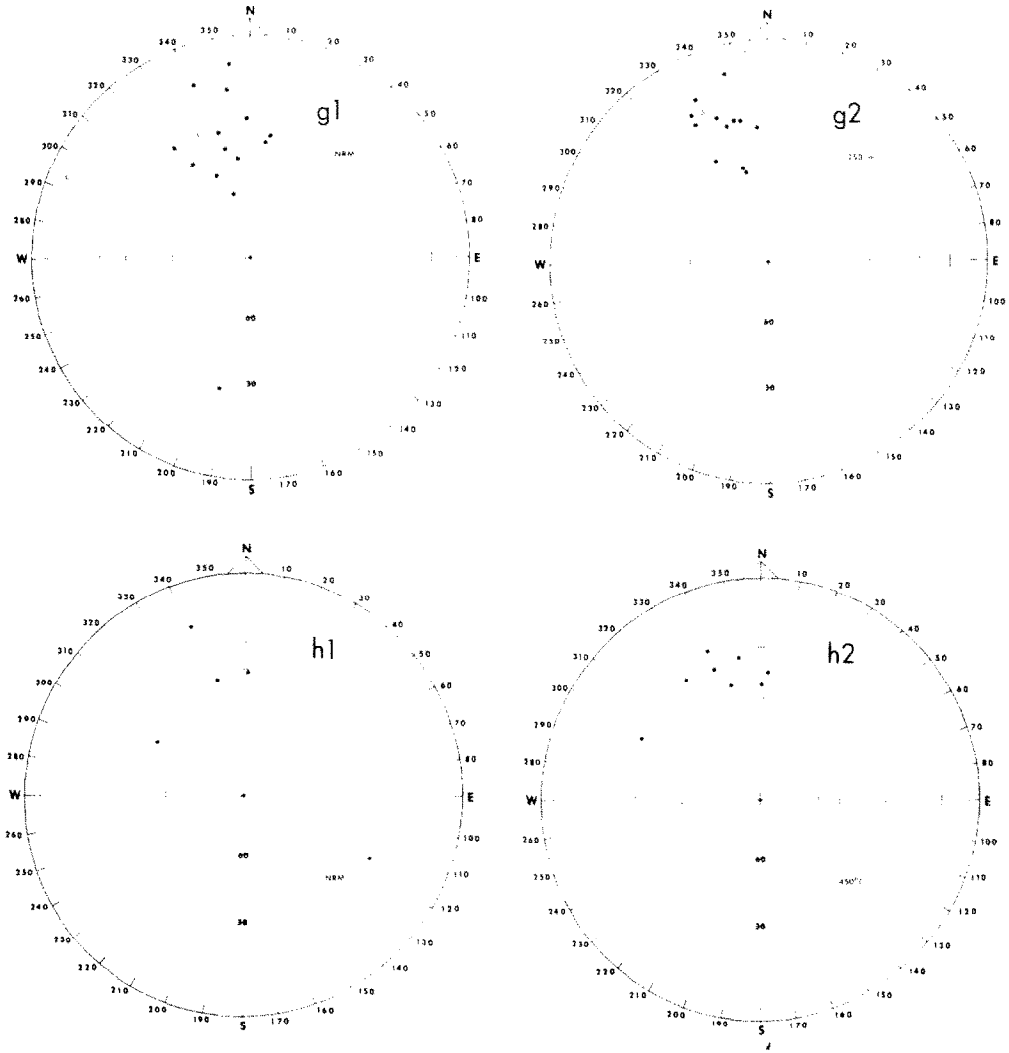


Fig. 3. Stereograms of the directions of magnetization of Moroccan samples. For each formation or group of results, the directions of natural remanent magnetization are shown in the left-hand figure (1). An * indicates some figures where fewer points after cleaning are the result of breakage in heating, or too low intensities of magnetization, or an occasion where NRM measurements were not made on all samples. a. Serie lie de vin, L. Cambrian. b. Various Carboniferous igneous rocks: quartz diorite, Tizi-n-Tichka, contact metamorphosed limestones, Bou Regreg pillow lava, and Bou Grou basalt. c. Upper Visean redbeds, Djebel Hadid. d. Visean limestones of Aftez. e. Djebel Tarhat Autunian redbeds. f. Argana redbeds from near Agadir. g. Liassic turbidites south of Midelt. h. Infra-Cenomanian sandstone samples from north of Ksar-es-Souk.

Carboniferous

Morocco is one of the few localities where both igneous and sedimentary rocks — acidic and basic intrusives and basic extrusives, redbeds and carbonates — may be collected for paleomagnetic work. In this study two basaltic flows, a granite and granite dyke from the Meseta (Fig. 1, *3*), the Tizi-n-Titchka massif (Fig. 1, *4*) in the High Atlas and the contact metamorphosed limestones provided the record from the igneous rocks. The sediments collected were from the upper (Visean (S_1—D_2)) redbeds of Djebel Hadid (Fig. 1, *5*), and Visean limestones of the Oued Draa and Tuisgui (Fig. 1, *6*) on the edge of the Sahara craton. Not all of those rocks yielded stable end points upon magnetic cleaning and for this reason work on the Tuisgui limestones and the acidic igneous rocks was abandoned.

One of the basaltic flows consisted of a pillow basalt exposed in the Oued Bou Regreg east of Rabat. It occurs within continental fanglomerates and interbedded shale with abundant plant remains which lie unconformably upon older sediments, probably Upper Devonian or lowermost Carboniferous. The second, sampled near Tiflet in the Oued Grou, consisted of a limited outcrop of basalt within a sequence of Lower Carboniferous flysch deposits (Tournaisian but possibly even Fammenian). Originally regarded as a dyke by earlier French workers, it is now considered contemporaneous with the flysch, for volcanic debris is concentrated in the sediments immediately above and below the flow horizon (Padgett, 1975).

The basalts were cleaned in an alternating field of 150 Oe based upon the results of the progressive demagnetization of pilot samples. These cleaned directions, and the stable end point from one sample are illustrated in Figs. 3b and 2b, the mean direction of magnetization and related statistical parameters are listed in Table I.

In the southern High Atlas, Walker and Vogel who recently described the Tizi-n-Titchka massif (Vogel and Walker, 1975) collected samples of the quartz diorite and contact metamorphosed limestones. The massif has been radiometrically dated, the K—Ar age is 297 ± 15 m.y. (Geochron), the Rb—Sr age is 294 m.y., (Termier et al., 1972). Measurements on two samples of the quartz diorite and three limestone samples collected by Walker and Vogel are illustrated in Fig. 3b and listed in Table I.

Upper Visean (S_1—D_2) redbeds collected on the slopes of Djebel Hadid belong to the Serie d'Izroutene described by Chanton-Guvenc et al. (1971). The sequence has recently been reinvestigated and remapped by Mullin (1974). The sequence, in excess of 200 m in thickness, consists of conglomeratic sandstone. The basal part is almost entirely conglomeratic, but in the upper part conglomeratic beds grade up into medium to coarse-grained sandstone. In outcrop the conglomerate levels form nearly vertical faces while the finer grained horizons from which the samples were obtained erode to form less-steep rubble-covered slopes. In consequence, since the identification of individual beds was often difficult, although recognition of

TABLE I

Summary of paleomagnetic data from various rock formations in Morocco. The paleopole positions are shown on Figs. 4a, b and c and may be identified from this listing number.

	Site location and formation		Age (m.y.)	Decl.	Incl.	α_{95}
PRECAMBRIAN AND CAMBRIAN						
—	Andesite lavas Bou Azzer	30.0N 7.0W	P.C. 620—1000	128.5	10.0	17
—	Ait Baha Amouslek Tuffs	30.0N 8.0W	L.C. 570—600	131.3	−19.6	13
M0	Serie lie de vin Anti-Atlas	30.5N 8.5W	L.C. 570—600 *	124.0	−1.7	20.5
* There is still debate as to whether this should be regarded as Cambrian or Precambrian.						
DEVONIAN						
H1	Norite south of Msissi	39.9N 4.7W	D. 350—360	315.5	−63.0	11.5
CARBONIFEROUS						
M4	Meseta Basalt flow	34.0N 6.0W	L. Carb.?	105.3	15.4	21.8
—	Tichka Massif diorite	31.0N 8.5W	330—340	113.8	28.0	
—	Tichka contact limestone			122.4	−12.1	21.6
M6	Bou Regreg pillow basalt	34.0N 7.0W	L. Carb.	116.4	23.8	12.7
M7	Djebel Hadid redbeds	33.5N 6.5W	L. Carb.	105.6	39.4	8.6
M8a	Oued Draa Aftez limestone	29.0N 9.0W	Visean	111.6	36.7	11.4
MC1	Moroccan Carboniferous geomagnetic pole, combination of all individual sites in M4, M6, M7,					
PERMIAN						
M10	Djebel Tarhat Redbeds	33.5N 6.5W	Autunian	120.6	3.6	7.8
H2	Chougrane Redbeds	33.0N 6.3W	Autunian	127.0	−6.0	4.7
H3	Taztot trachyandesite	32.2N 6.0W		136.0	−5.0	4.6
H4	Chougrane trachyandesite					
MC2	Moroccan Permian geomagnetic pole, combination of H2, H3, M10;					α_{95} = 12.1.
TRIASSIC						
M11	Teloutet lava	31.5N 7.4W	"Triassic"	316.2	44.2	16
M12	Argana Redbeds	30.5N 9.0W	U. Triassic	319.0	32.3	12.0
H5	Ait Aadel dolerites	31.6N 7.2W	M. Triassic	339.7	43.0	7.5
H6	Issaldain dolerites	31.1N 7.8W	M. Triassic	8.0	35.0	6.0

K	R	Sites	Samples	Treatment	Geomagnetic pole	Reference
11	7.374	8	43	—	30.6S 57.2E	Hailwood (1972)
9	14.340	16	87	—	41.5S 69.5E	Hailwood (1972)
8.2	7.151	8	8	450°C	29.2S 63.3E	This paper and Martin (1976)
29	—	8	42	—	0.5S 25.0E	Hailwood (1972)
13.1	4.696	1	5	150 Oe	8.1S 68.7E	This paper and Martin (1976)
	1.988	1	2		11.6S 55.9E	
33.5	2.940	1	3		30.8S 69.1E	
17.2	8.537	1	9	150 Oe	13.8S 57.2E	
26.4	11.583	12	12	500°C	0.1N 56.4E	
24.1	7.710	8	8	200 Oe	7.6N 52.5E	
M8a; α_{95} = 6.1.					4.8S 55.5E	
34.6	10.711	11	30	450°C	24.0S 63.8E	This paper and Martin (1976)
20		45	45		32.25S 64.1E	Daly and Pozzi (1976)
80		1	12		38.7S 56.8E	
					40.7S 54.0E	Hamzeh (1973) in Daly and Pozzi (1976)
					31.6S 61.7E	
24	4.382	1	5		51.4S 35.6E	This paper and Martin (1976)
7.5	19.231	22	22		50.6S 71.4E	
27	14.482	15	63		72.0S 74.5E	Hailwood (1975)
88	7.921	8	45		77.5S 39.5W	

TABLE I (continued)

Site	location and formation			Age (m.y.)	Decl.	Incl.	α_{95}
H7	Tizi 'n Tichka red sst.	31.5N	7.4W	M. Triassic	165.0	−23.0	22.0
H8	extra Sahara flows and intrusions			"Triassic"			22.0
H9	Jerada limestone	34.3N	2.2W	L. Triassic	347	54.6	4.0
H10	Khenifra basalt	32.9N	5.6W	M. Triassic	327.4	35.5	3.2
H11	Jerada Redbeds	34.3N	2.2W	L. Triassic	345	38.9	2.3
H12	Beni Snassen basalts	34.8N	2.1W	L. Triassic	326.3	37.6	7.8
H13	Tazzena basalts	34.1N	4.1W	L. Triassic	346.8	54.4	4.2
H14	Middle Atlas basalts			U. Triassic	347.0	31.0	7.0
MC3	Moroccan Triassic pole, combination of M11 and M12, H5, H6, H11, H12, H14; α_{95} = 15.8.						
JURASSIC							
M13a,b	Imiter massif (1)	33.0N	5.0W		185.1	14.3	26.0
	basalt flow (2)				339.5	3.5	23.8
M14	Limestone turbidites south of Midelt	35.7N	7.6W	L. Triassic	342.6	26.1	7.0
M15	Limestone outcrop near Rich	35.9N	7.6W	L. Triassic	342.7	68.0	7.1
M16	Imiter Massif gabbro	33.0N	5.0W	151±7—144±7	240.5	14.8	6.8
M17	Imiter Massif diorite	33.0N	5.0W		299.4	34.3	19.1
M18	Imiter Massif	33.0N	5.0W	165±10—162±10	265.5	26.1	20.6
M19	Bin-el Ouidane limestone	33.0N	5.5W	L. Jurassic	27.6	72.1	3.1
M20	Ouarouizarht limestone	33.0N	5.5W	L. Jurassic	29.7	51.5	9.8
H15	Draa Valley sills	29.8N	7.0W	181—187	339.0	27.5	4.5
H16	Foum Zguid dike	30.9N	6.9W	181—187	325.0	40.0	4.0
H17	Central Atlas Intrusives	31.5N	5.8W	181—187	318.5	38.5	22.0
mMCH	Moroccan Jurassic geomagnetic pole, combination of M14, H15;						α_{95} = 21.5.
CRETACEOUS							
M21	Red Sandstone near Kasr es Souk	32.0N	4.5W	"Infra-Cenoman."	340.8	29.8	15.5
H20	High Atlas Red Sandstone	31.7N	6.9W	"Infra-Cenoman."	347.5	39.0	6.0
H21	Red Sandstone near Beni Mellal			L. Cretaceous	314.0	28.0	8.0
mMC	Moroccan Cretaceous geomagnetic pole, combination of H20, H21;						α_{95} = 80.9.

K	R	Sites	Samples	Treatment	Geomagnetic pole	Reference
		4	22		68.5S 87.0E	
18		4	10		72.0S 81.0E	Bardon et al. (1973)
156		8	8		79.3S 96.0E	Daly and Pozzi (1976)
206		8	8		58.1S 68.0E	
201		19	19		72.0S 48.6E	
27		12	12		57.4S 72.0E	
185		6	6		79.1S 94.2E	
		190	190		71.0S 36.0E	Michard et al. (1975)
					70.8S 67.4E	
7.5	5.337	1	6	150 Oe	49.4S 12.8W	This paper and Martin (1976)
8.8	5.443	1	6	150 Oe	53.3S 30.8W	
42.8	10.766	11	11	250 Oe	61.6S 30.1E	
164.2	3.982	4	4	250 Oe	70.2S 138.8E	
66.2	7.894	1	8	150 Oe	19.7S 71.5E	
23.8	3.874	1	4	150 Oe	34.4S 86.2E	
11.4	5.565	1	6	150 Oe	3.7S 98.9E	
43.8	45.950	47	47	250 Oe	59.4S 156.0W	
9.5	22.487	25	25	250 Oe	72.5S 98.4W	
70	15.785	16	90		65.5S 50.5E	Hailwood and Mitchell (1971)
316	4.987	5	23		58.0S 79.0E	Hailwood and Mitchell (1971)
10	5.494	6	21		53.0S 81.5E	
					63.9S 39.5E	
15.9	6.624	7	7	450°C	66.3S 47.4E	This paper and Martin (1976)
117	4.966	5	28		75.0S 36.5E	Hailwood (1975)
		13	88		45.0S 71.0E	Michard et al. (1975)
					60.8S 61.9E	$\alpha_{95} = 80.9$

bedding strike and dip was not, samples have been treated together and a single mean calculated rather than a series of means from individual beds subsequently combined to provide an overall mean.

Stepwise demagnetization suggested that only a small secondary magnetic component was present (see Fig. 2c) and this appeared to be minimized at 500°C. The thermal cleaning of all samples at this temperature gave the lowest α_{95}—highest K-value. A stereogram of the results is given in Fig. 3c, with the appropriate parameters listed in Table I.

The most successful of the suites of limestone samples were the Visean limestones collected at Aftez (Fig. 1, 7) in the Oued Draa. These limestones were interbedded with calcareous sandstone horizons which behaved magnetically in a similar fashion so that no distinction is made between the two. The effects of demagnetization can be seen to result in a reduction of dip in the samples which gave a reasonable grouping of NRM values. It was also noted that the four scattered NRM measurements remained so after cleaning (see Fig. 3d). A stable end-point curve is illustrated in Fig. 2d. Normalized intensity curves indicate that in fields of 200 Oe only 40% of the NRM is retained. The results are illustrated in Fig. 3d and listed in Table I. The second suite of fossiliferous, micritic limestones collected from the limestone bed underlying the town of Tuisgui and a one-meter limestone sandwiched between orthoquartzitic offshore bars collected in a small oued near Messeid both gave dispersed results and did not reach a stable end point. These limestones were seen to be partially recrystallized in thin section (Martin, 1976).

Permian

Beds assigned to the Autunian are exposed on the slopes of Djebel Tarhat north of Khenifra (Fig. 1, 9). At outcrop, the beds present many similarities with the Lower Carboniferous deposits in the same general area. The sequence consists of conglomeratic units which grade upwards into more finely grained horizons. Exposures were located along the banks of a dry stream bed, and it was possible to sample from a number of distinct horizons. Consequently, after cleaning, results were combined to obtain mean directions for the unit as a whole. The samples were thermally cleaned, with the best result being that after cooling from 450°C. The cleaned data and a stable end-point curve are shown (Figs. 2e and 3e) with the relevant data listed in Table I.

Triassic

At four Triassic localities sampled in this study, two weathered basalt-flow sequences, one fresh basalt, and a section of red sediments were exposed. The two weathered basalt flows were too badly altered to obtain any useful data. The first was collected in a small spherodially weathered outcrop just

outside the town of Demnate, and the second came from a flow forming a resistant ridge near the road in the Argana Valley northeast of Agadir. The flow sequence in the scarp bordering the Argana Valley to the west was sampled by Bardon et al. (1973). The fresh basalt within a Triassic redbed sequence was collected in a small quarry just outside the town of Telouet (Fig. 1, *9*). This fresh lava yielded stable results which are shown in Table I.

The fourth Triassic locality collected was a sequence of redbeds well exposed in recent road cuts in the Argana Valley. Tixeront (1971) subdivided the Argana formation into eight sedimentary and one igneous members. Brown (1975) recently updated and refined that study showing that 2500—4500 m of clastic sedimentary redbeds were deposited in this area. The sediments derived from the east were deposited in a series of graben lows. The nature of the horst-and-graben topography accounts for the variation in the thickness of the section. The entire sequence is transgressional as it grades upward from continental fanglomerates to deltaics and is finally capped by offshore limestones. The samples collected in this study were taken mostly from the offshore prodelta clays and silts in the middle of the section (Brown, 1975). The redbeds were thermally demagnetized and showed least dispersion of residual directions at 450°C. The results are shown in Table I, and the stereogram (Fig. 3f) shows the results after treatment at 450°C. The figure also indicates the presence of reversely magnetized samples although the points are dispersed. Their stratigraphic distribution implies the presence of at least two reversed zones of relatively short duration.

Jurassic

A total of eight sites were collected in Jurassic intrusives, extrusives and sediments. The intrusives and extrusives (including two basalt flows) were all collected from the Imiter massif southwest of Midelt (Fig. 1, *11*). The first flow showed normal while the second showed reverse polarity. The results of these are tabulated (Table I) and shown on a stereogram (Fig. 3g). The best results upon AC cleaning were obtained at 150 Oe. Three other lithologies, a biotite gabbro dated by Geochron Laboratories at 144 ± 7 m.y., a hornblende diorite dated by Geochron at 162 ± 10 m.y., and a flow foliated diorite, were collected. These three will be discussed together because they behaved in a similar fashion. The site directions from these Imiter intrusives, all of which yielded good stable end-point curves, differ from all the class A data for the Jurassic of Africa. The rocks are fresh and relatively unaltered, leaving the most plausible interpretation as one involving local tectonic effects.

Four Liassic limestone localities were sampled in Morocco, all giving stable end-point curves. The first sequence was a series of unrecrystallized limestone turbidites which had been deposited off a shallow relief bank upon which pinnacle reefs were deposited. These were exposed in a road cut near the town of Rich, south of Midelt in the High Atlas mountains (Fig. 1, *12*).

After pilot samples were treated with AC fields up to 1000 Oe, it was determined that the least dispersion occurred in fields of 250 Oe. The results of all samples run at 250 Oe are tabulated in Table I and shown in a stereogram (Fig. 3g). A stable end-point curve (Fig. 2f) illustrates the stability of these samples. The data recovered after magnetic cleaning are reported in Table I.

Sixteen kilometers north of the turbidite section, a small outcrop of Jurassic limestone with a low dip was sampled. The dispersion was minimized in fields of 250 Oe. While the declinations of the turbidite section and this limestone section are the same, the inclination values of the latter are considerably higher. In thin section, the limestone was heavily recrystallized, and the simplest interpretation proposed is that this limestone acquired a stable secondary magnetic component at the time it was tilted into its present position.

Two Liassic (Lower Jurassic) carbonate sequences several hundred meters thick were collected near the town of Ouaouizarht, south of Beni Mellal (Fig. 1, *13*). Both sections went from subtidal limestone at the base to shallower-water sediments at the top and were capped by a continental redbed sequence. The series near the lake at Bin-el-Ouidane graded into the redbed sequence with carbonate cement persisting to higher levels in the section.

Pilot samples were partially demagnetized in fields up to 1000 Oe and the least dispersion was found at 250 Oe. The results of these two sequences are tabulated in Table I.

Thin-section examination revealed that both of these sections had undergone rather extensive recrystallization. The stable result recovered is therefore thought to partially or wholly represent a magnetization acquired at the time of recrystallization.

Cretaceous

A series of soft red infra-Cenomanian sandstones were sampled near the dam in road cuts on the road between Ksar Es Souk and Rich (Fig. 1, *14*). Eight sites were collected from this section and samples progressively demagnetized.

The results of all samples heated to 450°C are tabulated in Table I and shown in a stereogram (Fig. 3h). Figure 2g shows the stability end-point curve for one of the samples.

DISCUSSIONS AND RESULTS

Reliability of results

While it is normally considered that the stable magnetization remaining after magnetic cleaning represents the magnetization acquired at the time of formation of the rock or a geologically short time thereafter, Larson and Walker (1975) and Roy and Park (1974) have demonstrated convincingly

that in some redbeds this is not so. By implication, all redbeds are suspect and by extension, igneous rocks which have suffered alteration must also be treated with caution. It is therefore important that the virtual pole position from the Lower Carboniferous rocks, our principal addition to the Moroccan data, is demonstrably reliable.

The Tichka pluton (Vogel and Walker, 1975) is a shallowly emplaced mass formed from co-existing granitic and dioritic magma in which secondary alteration is negligible. Figure 3b shows that the directions of magnetization of the dioritic phase are similar to those found in the metamorphosed limestone contact zone. These directions are also the same as the two available basaltic lava flows from the Moroccan Meseta, the few limestone samples from African craton and the suite of redbed samples collected on the slopes of Djebel Hadid near Khenifra. This indicates that, if secondary, the magnetization of the sedimentary samples must have been acquired while the field had essentially the same orientation as at the time of emplacement of the igneous rocks, i.e. 240—320 m.y. (Choubert et al., 1965).

Similar arguments can be applied to sediments and several igneous rocks, when unaltered, for the Late Triassic and Jurassic. No such argument can be made for the Permian results, although these are assumed here to represent the Permian field which existed in Morocco.

Limestones, which were collected from horizons ranging from Devonian to Jurassic yielded mixed results. In some cases because of low initial intensities, close to the lower sensitivity level of the magnetometer, the intensities of magnetization are not reliable and so no stable end-point curves can be drawn. We have rejected suites where there were no stable end points. Three Liassic suites of limestones which yielded stable end points were rejected on the basis of strong recrystallization. A comparison of the uncrystallized turbidite limestones with recrystallized limestones some 16 km distant in the same section suggested that the magnetization of the latter may coincide with the time of tectonic activity.

Moroccan polar-wandering curve

As a means of summarizing the Moroccan data, both that presented here, and the work of other groups, a geomagnetic polar-wandering curve has been generated for Morocco (Fig. 4a). The data used in the computation have been identified and should be cross-checked with the total data listings of Tables I and II. It is stressed that these data fall into both the A and B classes of Hicken et al. (1972), largely depending upon the number of individually oriented samples.

For the purpose of incorporating Moroccan data with the African geomagnetic polar-wandering curve, however, only Hicken et al. (1972) class A data have been used. The Lower Carboniferous pole was obtained giving each stratigraphic horizon unit weight. In this computation the greater number of sites is from the red beds near Khenifra; however, the angular change to the pole position made by incorporating the rest of the data is only 5 degrees.

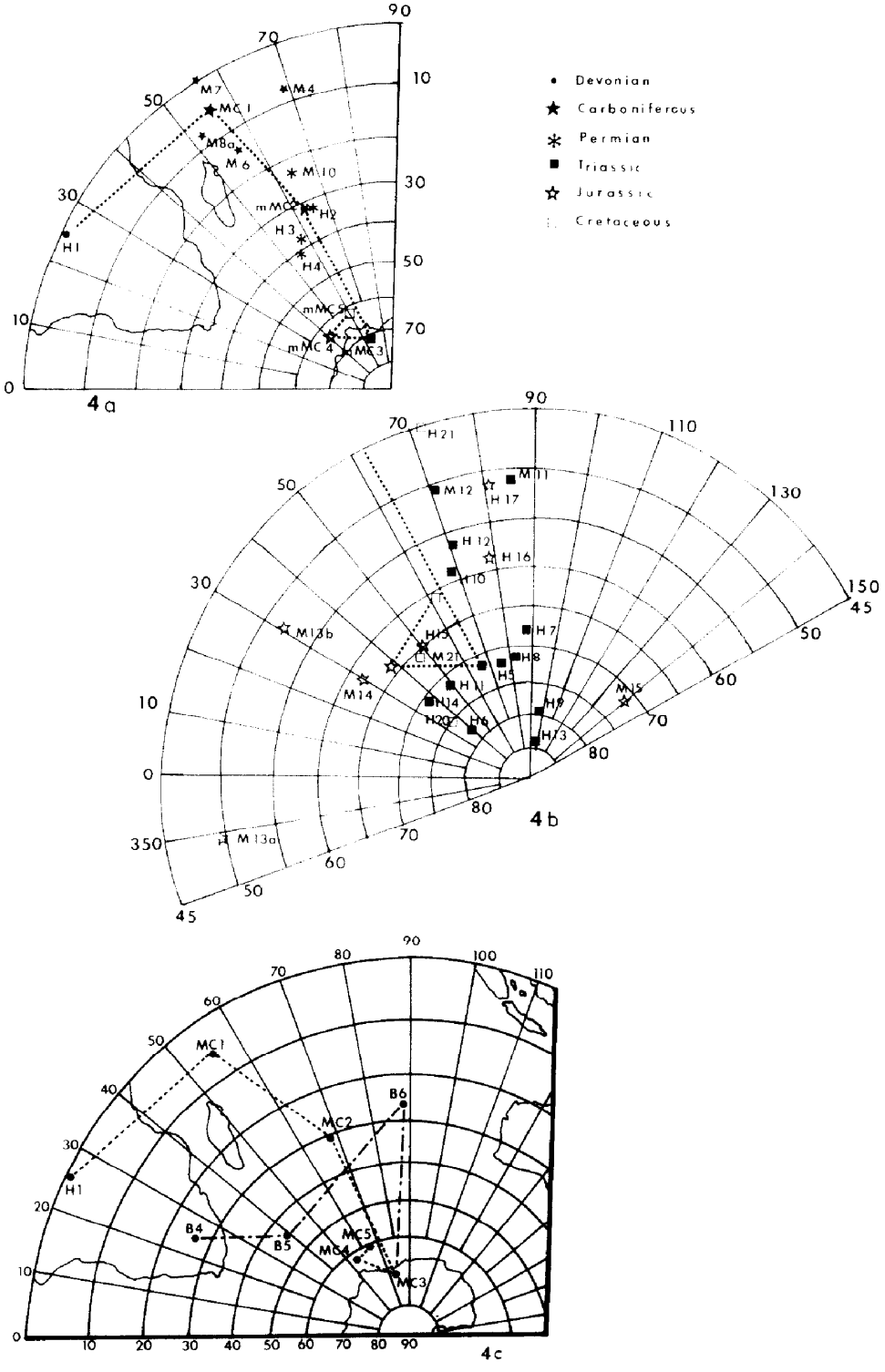

Devonian
Carboniferous
Permian
Triassic
Jurassic
Cretaceous
M7
M4
MC1
MBa
M6
M10
mMC
H2
H3
H4
H1
mMC5
mMC4
MC3
4a
H21
M12
M11
H17
H12
H16
H10
M13b
H15
M21
H8
H5
H7
H11
M14
H14
H20
H6
H9
M15
H13
M13a
4b
MC1
B6
MC2
H1
B4
B5
MC5
MC4
MC3
4c

TABLE II

Mean paleopole positions for Africa during the Late Paleozoic and Mesozoic. The data points used in the computation of the mean paleopole position are indicated. Each pole may be identified in Fig. 4b from its corresponding listing number. Only data which fall into class A category of Hicken et al. (1972) have been used in the computations

		Latitude	Longitude	α_{95}	k	Data used in computation [1]
H 1	Devonian	0.5S	25.0E	—	—	H1 [2]
Mc 1	Lower Carboniferous	4.8S	55.5E	—	—	(M4, M6, M7, M8a) [3]
Mc 2	Permian	31.0S	68.7E	15.7	34.8	H2, H3, M10, B6
Mc 3	Triassic	69.7S	75.3E	6.8	34.7	H5, H6, (M11 + M12) [4], H11, H12, B7, B8, B12, B11, B9, A1, A2, H14
Mc 4	Jurassic	64.6S	54.9E	11.8	61.1	M14, H15, A3, B10
Mc 5	Cretaceous	62.3S	65.2E	13.8	31.4	H20, H21, B15, B14, A4

[1] Data identified by letter in Table I with the following results from other parts of Africa:
B6 Ecca Redbeds, B7 Shawa Ijolite, B8 Triassic Red Sandstone Formation, B9 Maranudzi ring complex,
B11 Karroo dolerites, B12 Stormberg lavas, B13 Karroo lavas all from McElhinny et al. (1968),
A1 Freetown igneous complex (Briden et al., 1971), A2 Maji ya Chumvi Formation (McElhinny and Brock, 1975),
B10 Mateke Hills ring complex (McElhinny et al., 1968), A3 Liberian diabase dikes (Dalrymple et al., 1975),
B15 Lupata series volcanics (McElhinny et al., 1968), B14 Mlanje Massif syenite (McElhinny et al., 1968),
A4 Wadi Natash basalts (El Shazly and Krs, 1973).
[2] The only result from the Devonian of Africa though *not* class A data (only 8 sites).
[3] M8 is a combination of all Moroccan Carboniferous site poles, giving each geological horizon unit weight.
[4] (M11 + M12) The pole calculated from the single flow unit (M11) was combined with the 22 site poles represented in M12.

Integrity of Africa

The geomagnetic poles for Morocco during the Mesozoic correlate closely with those from the rest of Africa. However, the Permian virtual geomagnetic pole for Morocco, calculated from the result presented here and

Fig. 4. Late Paleozoic and Mesozoic paleopole positions from Africa. a. Paleozoic, Mesozoic paleopole positions from Moroccan rocks. Mean Moroccan Mesozoic paleopole positions are also shown for comparison. Paleopoles are identified by numbers corresponding to entries in Tables I and II. b. Mesozoic paleopole positions for African rocks. Triassic rocks are indicated by ■, Jurassic by ☆, Cretaceous by □. Each paleopole is numbered corresponding to entries in Tables I and II. c. African geomagnetic polar-wandering curve from Devonian to Triassic based upon class A data (Hicken et al., 1972) listed in Table II.

the data of Daly and Pozzi (1976) differs from that presented by Opdyke (1964), and it could be argued that this speaks in favor of relative motion of the Moroccan Meseta, from which the samples were collected, and to the rest of Africa at some time before the Triassic (Michard et al., 1975). However, there is no difference to be seen in the Carboniferous results from the Meseta, High-Atlas and Anti-Atlas, which appears to rule out relative motion, and in consequence provides a geomagnetic pole valid for the whole of Africa.

If this is so, there is a problem in reconciling the Lower Carboniferous and Permian (Autunian) data from Morocco with the Lower and Upper Carboniferous and Permian data from Central Africa (points B4,5,6; McElhinny et al., 1968).

It can be seen from Fig. 4 that the Moroccan Permian geomagnetic pole position is about the same as the mean of points B5 and B6 (Upper Carboniferous(?) and Lower Permian), but that the B4 result is much closer to Hailwood's Devonian geomagnetic pole than to the Lower Carboniferous geomagnetic pole reported here. McElhinny and Opdyke (1968), in the absence of comparative African data, argued that by analogy with Australia the Dwyka result, and thus the glacial episode, occurred during Early Carboniferous or even in the late Devonian times and quote the identification of *Leptophloem australe* found in the Orange Free State in shales interbedded in the tillite directly resting upon Dwyka tillite by Plumstead as independent support of this proposal. On the other hand, the discovery in South West Africa of the goniatite *Eosianites* subgenus *Glaphyrites* (Martin et al., 1972) closely related to Uruguayan forms suggests a Late Carboniferous age. If it were possible that the Dwyka glacial sediments did indeed extend from Late Devonian to Early Carboniferous, though not necessarily everywhere the same age, the problem could be resolved — although the need for good Devonian data becomes critical.

African polar-wandering curve

The principal new result presented here concerns the Lower Carboniferous just discussed. We have also modified the location of the Permian virtual pole position by averaging the Moroccan and southern African results (Fig. 4c). In consequence, the late Paleozoic segment of the African polar-wandering curve deviates somewhat from that presented by McElhinny (1968), while the Mesozoic segment is essentially unaltered.

Irving (1964) suggested a period of rapid polar wandering during Late Paleozoic time after he called attention to paleomagnetic data indicating a rapid change in the latitude of southeastern Australia during the Carboniferous. Irving (1966) found that the Lower Carboniferous rocks of southeastern Australia have intermediate magnetic inclinations while the Upper Carboniferous rocks have steep magnetic inclinations consistent with a relative polar shift of about 40° in latitude during Namurian—Westphalian time. He termed

this interval of rapid polar translation a transition. The data presented here clearly lend support to rapid translation during the Carboniferous, however the data are interpreted.

The concept of long time intervals characterized by slow polar wandering separated by shorter intervals of more rapid polar motion was extended by Briden (1967a) to all Gondwana continents and McElhinny and Opdyke (1968), by assuming an Early Carboniferous to Late Devonian age for the Central African Dwyka varved sediments, determined a 55° movement of the pole relative to Africa during Carboniferous time. Creer (1970) reviewed the paleomagnetic data through 1969 and concluded that a shift of the poles took place between the Early and Late Carboniferous in Europe and that the data from other continents were consistent with this interpretation. Tarling (1971) considered that Paleozoic paleomagnetic data were too poorly defined for much resolution of rapid polar motions, although there was reason to accept the estimates of polar-wander rates during the periods of slow polar movement. McElhinny (1973) refutes a Middle Carboniferous transition interval for Africa but retains a Gondwana stable interval in the Late Carboniferous to Early Permian with the implication that this was preceded by a transition interval.

African paleomagnetic data through 1977 indicate a large relative movement of the continent relative to the pole between Devonian and Triassic time, but the average rate is 1° per million years, which is less than the rate of sea-floor spreading of the Indian Ocean. The results presented here are from a comprehensive sampling of Lower Carboniferous basalt flows, pillow lavas, redbeds and limestones, and Lower Permian (Autunian) redbeds. These results indicate an essentially eastward movement for northwest Africa of 26° over an 85 m.y. interval of the Permo-Carboniferous, averaging 0.30° per million years. These results serve to modify the Late Paleozoic segment of the African polar-wandering curve as presented by McElhinny et al. (1968), but the rate of polar wandering for the Permo-Carboniferous is similar to the rate derived from Pennsylvanian coal paleomagnetic data recently obtained by Noltimier and Kopacz (1975) and Kopacz and Noltimier (1976).

There are two ways in which the polar-wandering curve for the Late Paleozoic may be drawn, a relatively smooth curve from Devonian through the new Early Carboniferous VGP location to the revised Permian VGP as has been done here, or to introduce a large dog-leg from Early Carboniferous to the Late Carboniferous VGP position given by McElhinny, a choice which involves a greater rate of polar wandering but retains the idea of a mid-Carboniferous transition. We have chosen the former since we suspect that there may be a stable secondary component affecting the Upper Carboniferous in the manner indicated by Larson and Walker (1975), but the second choice remains a possibility.

The close grouping of the Mesozoic VGP noted by McElhinny (1968) has been augmented by new VGP determinations. This suggested that an attempt should be made to see if small shifts are detectable. The data are from rocks

which cover a relatively small time interval, most within the Late Triassic—Early Jurassic. Taking the stratigraphic information provided for Upper Triassic, Triassic—Jurassic boundary, and Lower Jurassic intervals, it was impossible to discriminate between them. However, grouping Triassic (MC3, Fig. 4c) Jurassic (s. l.) (MC4, Fig. 4c) and Cretaceous (MC5, Fig. 4c) (Table II) a small loop is apparent, but since the confidence limits of the mean poles overlap, it is premature to assign great significance to it.

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REFERENCES

Bardon, C., Bossert, A., Hamzeh, R., Rolley, J.P. and Westphal, M., 1973a. Etude Paléomagnétique de Formations Volcaniques du Crétace Inférieur dans l'Atlas de Beni Mellal (Maroc). C. R. Acad. Sci. Paris, 227: 2141—2144.

Bardon, C., Bossert, A., Hamzeh, R. and Westphal, M., 1973b. Etude Paléomagnetique de Formations du Trias et du Jurassique du Maroc et du Sahara. C. R. Acad. Sci. Paris, 276: 2356—2360.

Briden, J.C., 1967a. A new paleomagnetic result from Lower Cretaceous of East-Central Africa, Geophys. J., 12: 375—380.

Briden, J.C., 1967b. Recurrent continental drift of Gondwanaland. Nature, 215: 1334—1339.

Briden, J.C., Henthorn, D.J. and Rex, D.C., 1971. Palaeomagnetic and radiometric evidence for the age of the Freetown Igneous complex, Sierra Leone. Earth Planet. Sci. Lett., 12: 385—391.

Brown, R., 1975. The Argana Basin of Morocco: A Triassic Model for Early Rifting. Unpubl. M.S. Thesis, Univ. of South Carolina.

Chanton-Guvenc, N., Huvelin, P. and Semenoff-Trian-Charsky, P., 1971. Les deux séries d'age Visean Supérieur du Jbel Hadid près de Khenifra (Maroc Hercynien Central). Notes Serv. Géol. Maroc., 31: 7—10.

Choubert, G., Diouri, M. and Faure-Muret, A., 1965. Mesures géochronologiques récentes par la méthode A^{40}/K^{40} au Maroc. Notes Serv. Géol. Maroc, 24: 5362.

Choubert, G., Faure-Muret, A. and Hottinger, L., 1971. La série stratigraphique de Tarfaya (Maroc Méridional) et le problème de la naissance de l'océan Atlantique, Notes Serv. Géol. Maroc, 31: 29—40.

Creer, K.M., 1970. A review of paleomagnetism. Earth Sci. Rev., 6: 369—466.

Dalrymple, G.B., Gromme, S.C. and White, R.W., 1975. Potassium—argon age and paleomagnetism of diabase dikes in Liberia: Initiation of Central Atlantic rifting. Geol. Soc. Am. Bull., 86: 399—411.

Daly, L. and Pozzi, J-P., 1976. Résultats paléomagnétiques du permien inférieur et du trias Marocain; Comparaison avec les données Africaines et Sud Américaines. Earth Planet. Sci. Lett., 29: 71—80.

El Shazly, E.M. and Krs, M., 1973. Paleogeography and paleomagnetism of the Nubian Sandstone eastern desert of Egypt. Geol. Rundsch., 62: 212—225.

Hailwood, E.A., 1972. Paleomagnetic studies on rock formation in the High Atlas and Anti-Atlas regions of Morocco. Unpubl. Ph.D. thesis, Univ. of Newcastle upon Tyne.

Hailwood, E.A., 1974. Paleomagnetism of the Msissi norite (Morocco) and the Paleozoic reconstruction of Gondwanaland. Earth Planet. Sci. Lett., 23: 376—386.

Hailwood, E.A., 1975. The paleomagnetism of Triassic and Cretaceous rocks from Morocco. Geophys. J., 41: 219—235.

Hailwood, E.A. and Mitchell, J.G., 1971. Paleomagnetic and radiometric dating results from Jurassic intrusions in south Morocco. Geophys. J., 24: 351—364.

Hamzeh, R., 1973. Thèse de spécialité. Strasbourg, unpublished.

Helsley, C.E., 1965. Paleomagnetic results from the Middle Cambrium of Northwest Africa. Trans. Am. Geophys. Union, 46: 67 (abstract).

Hicken, A., Irving, E., Law, L.H. and Hastie, J., 1972. Catalogue of Paleomagnetic Direction and Poles, vol. 45, No. 1. Dept. of Energy, Mines and Resources, Ottawa, Canada.

Hurley, R.M. and Rand, J.R., 1969. Pro-drift continental nuclei. Science, 164: 1229—1242.

Irving, E., 1964. Paleomagnetism and its Applications to Geological and Geophysical Problems. Wiley, New York, 399 pp.

Irving, E., 1966. Paleomagnetism of some Carboniferous rocks from New South Wales and its relation to geological events. Geophys. J., 71: 6025—6051.

Kopacz, M.A. and Noltimier, H.C., 1976. The coal pole and the Kittanning reversal. Trans. Am. Geophys. Union, 57: 238 (abstract).

Larson, E.E. and Walker, T.R., 1975. Development of chemical remanent magnetization during early stages of redbed formation in late Cenozoic sediments. Geol. Soc. Am. Bull., 86: 639—650.

Martin, D.L., 1976. Paleomagnetic Results from Morocco, New Results and Survey, and a Detailed Study of the Paleomagnetics of Carbonates. Unpubl. Ph.D. thesis, Univ. of South Carolina.

Martin, H., Walliser, O.H. and Wilczewski, N., 1972. A goniatite from the Glaciomarine Dwyka beds near Schlip, South West Africa. Proc. 2nd Gondwana Conf., pp. 621—625.

McElhinny, M.W., 1968. The mid-Carboniferous continental drift episode — evidence from Australia and Africa. In: IVGS, UNESCO symposium on Continental Drift, Montevideo.

McElhinny, M.W., 1973. Palaeomagnetism and Plate Tectonics. Cambridge Univ. Press, London, 201 pp.

McElhinny, M.W. and Brock, A., 1975. A new paleomagnetic result from East Africa and estimates of the Mesozoic paleoradius. Earth Planet. Sci. Lett., 27: 321—328.

McElhinny, M.W. and Opdyke, N.D., 1968. Paleomagnetism of some Carboniferous glacial varves from Central Africa. J. Geophys. Res., 73: 689—696.

McElhinny, M.W., Briden, J.C., Jones, D.L. and Brock, A., 1968. Geological and geophysical implications of paleomagnetic results from Africa. Rev. Geophys., 6: 201—238.

Michard, A., Westphal, M., Bossert, A. and Hamzeh, R., 1975. Tectonique de blocs dans le socle Atlaso-Mesetien du Maroc: Une nouvelle interprétation des données géologiques et paléomagnétiques. Earth Planet. Sci. Lett., 24: 363—368.

Mullin, P.R., 1977. Stratigraphy, Structure and Sedimentation of the Carboniferous Rocks of the Khenifira Region, Central Morocco. Unpubl. Ph.D. thesis, University College, Swansea.

Nairn, A.E.M., 1975. Germanotype Tektonik und die Plattentektonik-hypothese. Geol. Rundsch., 64: 716—727.

Noltimier, H.C. and Kopacz, M.A., 1975. The coal pole. Trans. Am. Geophys. Union, 56: 354 (abstract).

Opdyke, N.D., 1964. The paleomagnetism of the Permian redbeds of southwest Tanganyika. J. Geophys. Res., 69: 2477—2487.

Padget, G., 1975. Sedimentary Analysis of the Rabat Basin, Morocco. Unpubl. M.S. thesis, Univ. of South Carolina.

Piper, J.D.A., 1973. Geological interpretation of paleomagnetic results from the African Precambrian. In: D.H. Tarling and S.K. Runcorn (Editors), Implication of Continental Drift to the Earth Sciences. Academic Press, London, vol. 1: 19—21.

Roy, J.L. and Park, J.H., 1974. The magnetization process of certain redbeds: Vector analysis of chemical and thermal results. Can. J. Earth Sci., 11: 437—471.

Spall, H., 1972. Did southern Africa and North America drift independently during the Precambrian? Nature, 236: 219—221.

Tarling, D.H., 1971. Gondwanaland, palaeomagnetism and continental drift. Nature, 229: 17—21.

Tarling, D.H. and Sutton, D.H., 1967. Paleomagnetic stability results of a reconnaissance survey of red sandstones in Morocco. Phys. Earth Planet. Inter., 1: 35—43.

Termier, H., Termier, G., Leutweein, F. and Sonet, J., 1972. Sur l'age Varistique des plutonites du Massif de Tichka (Maroc). Bull. Soc. Fr. Min. Cristallogr., 95: 124—127.

Tixeront, M., 1971. Lithostratigraphie et minéralisations cuprifères et uranifères stratiformes syngenétiques et familières des formations détritiques du couloir d'Argana (Haut Atlas Occidental Maroc) et possibilités de récherches. Rapp. S.E.G.M., 921: 28.

Vogel, A. and Walker, B.M., 1975. The Tichka massif, Morocco — an example of contemporaneous acidic and basic plutonism. Lithos, 8: 29—38.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: Analysis of results. In: Collinson, Creer, and Runcorn (editors), Methods in Palaeomagnetism, Vol. 3: 254—286.